30-Day Readmissions by Chronic Condition Profile

A HEDIS inspired measure built in BigQuery on synthetic Medicare claims

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Members with diabetes, kidney disease, and heart failure had a 15.6% 30-day readmission rate, compared with 4.0% among members with none of these conditions. They accounted for 36% of eligible stays and 57% of readmissions, making it a useful starting point for outreach and case management efforts aimed at reducing readmissions.

What the data shows

Member health profile	Eligible stays	Readmission rate
All 3 conditions	23,383	15.6%
Kidney & heart failure	2,531	11.6%
Diabetes & kidney disease	3,815	9.1%
Diabetes & heart failure	11,118	8.4%
Kidney disease only	1,072	7.8%
Heart failure only	4,269	6.0%
Diabetes only	8,149	5.6%
No tracked conditions	9,961	4.0%
All members	64,298	10.0%

Readmission rates generally increased with the number of tracked chronic conditions. Kidney disease was present in the three highest-rate groups, although the kidney disease only group had a relatively small sample of 1,072 stays. These patterns identify areas for further investigation but do not establish that any particular condition causes readmissions.

How I built it

I counted a stay as a readmission if the same member was readmitted 1 to 30 days after discharge. I used a window function (LEAD) in BigQuery to find each member's next admission. Of 66,754 stays, 64,298 were eligible, and 6,423 of those were readmissions.

I removed 19 duplicate rows, 313 stays discharged too close to the end of the data to have a full 30 days, and 2,143 stays (3.2%) where the next admission began on or before the discharge date, which looks like overlapping billing. The profile counts add up to the overall total, and no stay has a discharge before its admission.

Limits

The dataset is synthetic, so the results demonstrate the analytical method rather than real-world readmission patterns. The measure is modeled on HEDIS but is not an exact reproduction. The analysis does not distinguish planned from unplanned readmissions, adjust for patient risk, or identify transfers. Stays from members absent from the 2008 enrollment file were excluded, and a decline in claim volume late in 2010 may result in missed readmissions near the end of the observation period.